RC CIRCULAR COLUMN CHECK – FLEXURE AND SHEAR (EC2)

1. DESIGN BASIS

EN 1992-1-1. Strips and bars, strain pivots A/B/C, P–M curve in the MEd direction.

Flexure: ray from 0 through (MEd, NEd) to the P–M curve (UR_M). fcd and shear (spirals):

EN 1992-1-1:2023 (5.3), §8.2.3.

N positive in compression. Annex A.n is calculated at its link ▸.

2. INPUTS

2.1 Materials

Characteristic concrete cylinder strength $f_{ck} := 35 \cdot \mathrm{MPa}$

Partial factor for concrete $\gamma_C := 1.5$

Sustained-load factor (5.3): 0.85, or 1.0 if loaded after 90 days $k_{tc} := 0.85$

Yield strength of longitudinal bars $f_{yk} := 500 \cdot \mathrm{MPa}$

Yield strength of spirals $f_{ywk} := 500 \cdot \mathrm{MPa}$

Ductility class of the spirals: 1 = class A, 0 = class B or C $\delta_A := 0$

Partial factor for reinforcement $\gamma_S := 1.15$

Elastic modulus of reinforcement $E_s := 200 \cdot \mathrm{GPa}$

Ultimate concrete strain (Table 3.1) $\varepsilon_{cu2} := 0.0035$

Concrete strain at peak stress $\varepsilon_{c2} := 0.002$

Design limit of steel strain, 0.9·εuk $\varepsilon_{ud} := 0.045$

2.2 Section and reinforcement sets

Column diameter $D := 1200 \cdot \mathrm{mm}$

Nominal cover to the set-1 spiral $c_{nom} := 75 \cdot \text{mm}$

Reinforcement sets: 1 = outer, 2 = outer + inner $n_{set} := 1$

Set 1 (outer): longitudinal bar diameter $\phi_{l1} := 32 \cdot \text{mm}$

Set 1: number of bars (evenly spaced) $n_{l1} := 22$

Set 1: spiral bar diameter $\phi_{w1} := 20 \cdot \text{mm}$

Set 1: spiral pitch $s_1 := 85 \cdot \text{mm}$

Set 2 (inner): longitudinal bar diameter $\phi_{l2} := 25 \cdot \text{mm}$

Set 2: number of longitudinal bars $n_{l2} := 16$

Set 2: spiral bar diameter $\phi_{w2} := 16 \cdot \text{mm}$

Set 2: spiral pitch $s_2 := 150 \cdot \text{mm}$

2.3 Integration

Number of concrete strips across D $n_c := 100$

2.4 Design actions (ULS)

Axial force, compression positive $N_{Ed} := 8000 \cdot \text{kN}$

Bending moment about x $M_{Ed.x} := 3000 \cdot \text{kN} \cdot \text{m}$

Bending moment about y $M_{Ed.y} := 2000 \cdot \text{kN} \cdot \text{m}$

Shear force $V_{Ed} := 2500 \cdot \text{kN}$

3. DESIGN VALUES

Effective strength factor (5.4), fck,ref = 40 MPa $\eta_{cc} := \min\left(\left(\frac{40 \cdot \text{MPa}}{f_{ck}}\right)^{\frac{1}{3}}, 1\right) = 1$

Design compressive strength (5.3) $f_{cd} := \frac{\eta_{cc} k_{tc} f_{ck}}{\gamma_C} = 19.83 \cdot \text{MPa}$

Design yield strength of longitudinal bars $f_{yd} := \frac{f_{yk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strength of spirals $f_{ywd} := \frac{f_{ywk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strain of longitudinal bars $\varepsilon_{yd} := \frac{f_{yd}}{E_s} = 0.00217391$

Set-2 switch: 1 = included, 0 = ignored

$$\delta_2 := \begin{cases} 0 & \text{if } n_{set} \geq 2 \\ 1 & \text{else} \end{cases}$$

4. SECTION GEOMETRY

4.1 Rings, bars and spirals

Section radius R (centroid to the outer face) $R := \frac{D}{2} = 600 \cdot \text{mm}$

Cover to set-2 spiral (inside set-1 bars) $c_2 := c_{nom} + \phi_{w1} + \phi_{l1} = 127 \cdot \text{mm}$

Radius to set-1 bar centres $r_1 := R - c_{nom} - \phi_{w1} - \frac{\phi_{l1}}{2} = 489 \cdot \text{mm}$

Radius to set-2 bar centres $r_2 := R - c_2 - \phi_{w2} - \frac{\phi_{l2}}{2} = 444 \cdot \text{mm}$

Set 2 must sit inside set 1 $r_2 = 444 \cdot \text{mm} \leq r_1 - \frac{\phi_{l1} + \phi_{l2}}{2} = 460 \cdot \text{mm}$ PASS

Area of one set-1 longitudinal bar $A_{b1} := 0.25\pi\phi_{l1}^2 = 804 \cdot \text{mm}^2$

Area of one set-2 longitudinal bar $A_{b2} := 0.25\pi\phi_{l2}^2 = 491 \cdot \text{mm}^2$

Gross section area $A_g := \pi R^2 = 1130973 \cdot \text{mm}^2$

Total longitudinal steel area $A_{st} := n_{l1} A_{b1} + \delta_2 n_{l2} A_{b2} = 17693 \cdot \text{mm}^2$

Spiral centreline diameter, set 1 $D_{sp.1} := D - 2c_{nom} - \phi_{w1} = 1030 \cdot \text{mm}$

Spiral centreline diameter, set 2 $D_{sp.2} := D - 2c_2 - \phi_{w2} = 930 \cdot \text{mm}$

Area of one set-1 spiral bar $A_{w1} := 0.25\pi\phi_{w1}^2 = 314 \cdot \text{mm}^2$

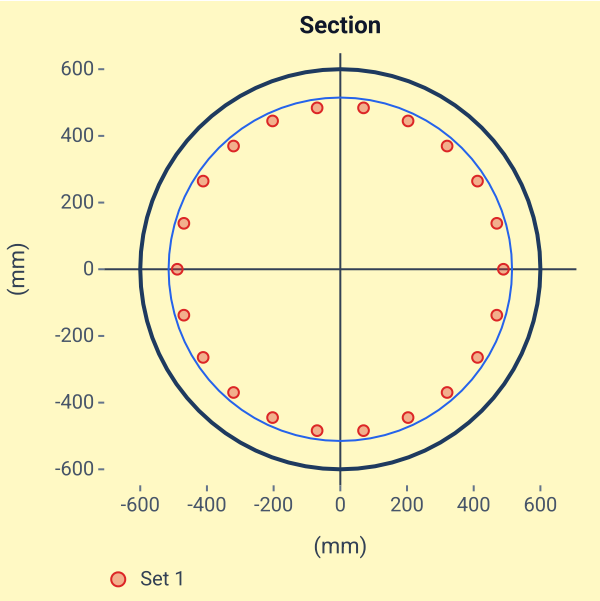
Area of one set-2 spiral bar $A_{w2} := 0.25\pi\phi_{w2}^2 = 201 \cdot \text{mm}^2$

Clear spacing of set-1 bars on the ring $s_{cl.1} := \frac{2\pi r_1}{n_{l1}} - \phi_{l1} = 108 \cdot \text{mm}$

Minimum clear spacing, EN 1536 bored piles $s_{\text{cl.1}} = 108 \cdot \text{mm} \geq 100 \cdot \text{mm}$ PASS

4.2 Bars and section

▸ **A.1** Bar coordinates · see page 9



4.3 Concrete strips

▸ **A.2** Concrete strips · see page 10

5. INTERACTION CURVE (N–M AT THE DEMAND DIRECTION)

5.1 Bending direction

Resultant design moment $M_{Ed} := \sqrt{M_{Ed.x}^2 + M_{Ed.y}^2} = 3605.6 \cdot \text{kN} \cdot \text{m}$

Direction of MEd, signs kept (0 ... 360°) $\theta_M := \text{angle}\left(M_{Ed.x}, M_{Ed.y}\right) = 33.7^\circ$

▸ **A.3** Strain plane, stress laws and section resultants · see page 11

5.2 Control points of the curve

The curve is evaluated at 55 neutral-axis depths from pure tension to uniform compression.

▸ **A.4** P–M diagram data (control points) · see page 12

Max. axial resistance (uniform εc2) $N_{Rd.max} = 29159.2 \cdot \text{kN}$

Maximum moment resistance on the curve $M_{Rd.max} = 4726.7 \cdot \text{kN} \cdot \text{m}$

6. FLEXURAL CHECK

Ray from the origin through the design point (MEd, NEd) to the P–M curve, as MBColumn.

6.1 Radial ratio

Normalised design axial force $n_{Ed} := \frac{N_{Ed}}{N_{Rd.max}} = 0.27436$

Normalised design moment $m_{Ed} := \frac{M_{Ed}}{M_{Rd.max}} = 0.7628$

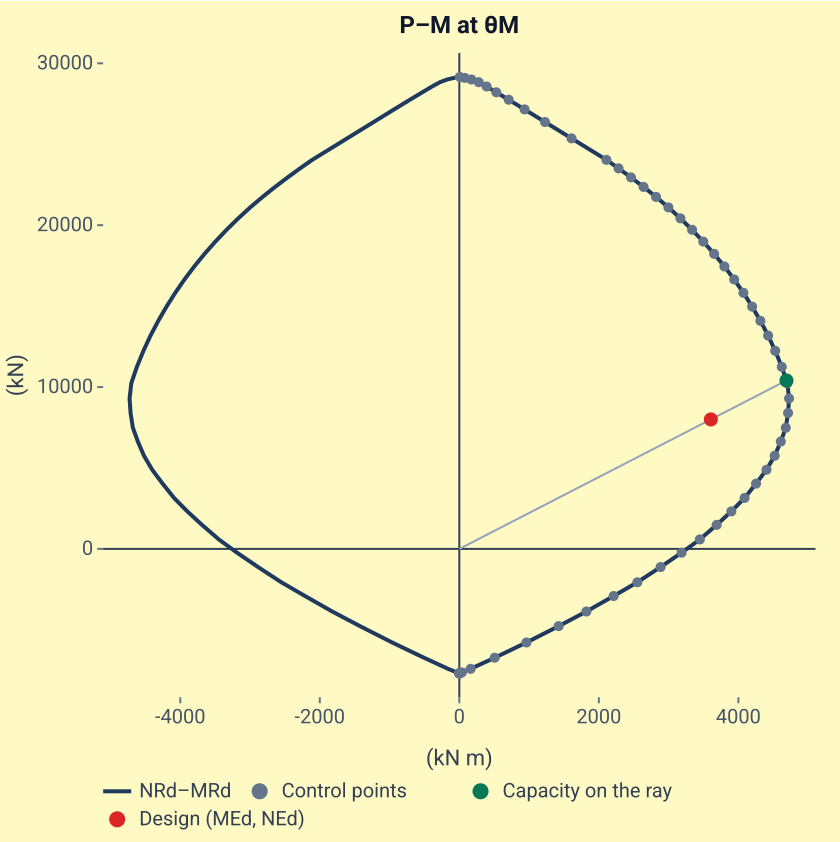
Distance of the design point from the origin $\rho_{Ed} := \sqrt{n_{Ed}^2 + m_{Ed}^2} = 0.81064$

Curve distance along the design ray $\rho_{Rd} := \text{linterp}\left(s_v, \rho_v, \frac{n_{Ed}}{\rho_{Ed}}\right) = 1.0543$

Radial ratio (scale-independent) $\text{UR}_M := \frac{\rho_{Ed}}{\rho_{Rd}} = 0.7689$

Check $\mathrm{UR}_M = 0.7689 \leq 1$ PASS

6.2 P–M diagram at θM



7. SHEAR CHECK (EN 1992-1-1:2023 §8.2.3, SPIRALS)

7.1 Chords, web width and stresses

Truss in stresses: τEd = VEd/(bw·z) (8.18); spirals τRd,sy = ρw·fywd·cotθ (8.42); strut σcd = τEd·(cotθ + tanθ) ≤ ν·fcd (8.44), ν = 0.5 (8.2.3(6)). bw: least width between the compression chord (centroid of the concrete compression, §8.1) and the tension chord (d). Each spiral crosses the shear plane twice: Asw = 2·Aw.

Effective depth: D/2 + half-ring centroid 2r/π, weighted by bar area

$$d := \frac{D}{2} + \frac{2}{\pi} \cdot \frac{n_{l1} A_{b1} r_1 + \delta_2 n_{l2} A_{b2} r_2}{n_{l1} A_{b1} + \delta_2 n_{l2} A_{b2}} = 911 \cdot \text{mm}$$

Lever arm, 8.2.1(3) $z := 0.9d = 820 \cdot \text{mm}$

Neutral-axis depth at the bending resistance under NEd (§8.1) $x := \text{linterp}\left(N_v, c_v, N_{Ed}\right) = 587 \cdot \text{mm}$

Depth of the compression chord centroid (concrete compression at x)

$$y_{cc} := \left\| \begin{array}{l} \text{if } x \le 0 \cdot \text{mm} \\ \quad \| 0 \cdot \text{mm} \\ \text{else} \\ \quad \left\| R - \frac{\text{sum}\left(\sigma_c\left(\varepsilon\left(x, x_c\right)\right) A_c x_c\right)}{\text{sum}\left(\sigma_c\left(\varepsilon\left(x, x_c\right)\right) A_c\right)} \right. \end{array} \right. = 286 \cdot \text{mm}$$

Web width: least chord width between the chords (only the tension chord if x ≤ 0)

$$b_w := \left\| \begin{array}{l} \text{if } x \le 0 \cdot \text{mm} \\ \quad \left\| 2\sqrt{R^2 - (d - R)^2} \right. \\ \text{else} \\ \quad \left\| \min\left(2\sqrt{R^2 - (R - y_{cc})^2}, 2\sqrt{R^2 - (d - R)^2}\right) \right. \end{array} \right. = 1022 \cdot \text{mm}$$

$$\tau_{Ed} := \frac{V_{Ed}}{b_w z} = 2.98 \cdot \text{MPa}$$

Average shear stress (8.18)

Strength reduction for cracked concrete, 8.2.3(6) $\nu := 0.5$

7.2 Strut angle

1 ≤ cotθ ≤ cot θmin (8.41): 2.5; 3.0 if σcp ≥ 3 MPa and x < 0.25d (linear between); 2.5 if x ≥ 0.25d; 2.5 − 0.1·|NEd|/VEd ≥ 1 in tension; × kA = 0.8 for class A spirals. cotθ < 1 (8.2.3(11)) is not used: σcd is least at θ = 45° and τRd,sy falls with cotθ.

Design cotθ: the largest value ≤ cot θmin that keeps σcd ≤ ν·fcd.

Factor on cot θmin: 0.8 for class A spirals, 8.2.3(4) $k_A := 1 - 0.2\delta_A = 1$

$$\sigma_{cp} := \frac{N_{Ed}}{A_g} = 7.07 \cdot \text{MPa}$$

Average axial stress, compression positive

Flattest strut allowed (8.41)

$$\cot_{\theta min} := \begin{cases} \text{if } N_{Ed} < 0 \cdot \text{kN} \\ \quad \max\left(1, k_A \left(2.5 - 0.1 \cdot \frac{|N_{Ed}|}{V_{Ed}}\right)\right) \\ \text{else if } x \geq 0.25d \\ \quad 2.5k_A \\ \text{else if } \sigma_{cp} \geq 3 \cdot \text{MPa} \\ \quad 3.0k_A \\ \text{else} \\ \quad \left(2.5 + 0.5 \cdot \frac{\sigma_{cp}}{3 \cdot \text{MPa}}\right) k_A \end{cases} = 2.5$$

Design cotθ: cot θmin if the strut holds, else the root of τEd(cotθ + tanθ) = ν·fcd; 1: strut fails

$$\cot_\theta := \begin{cases} \text{if } \tau_{Ed}\left(\cot_{\theta min} + \frac{1}{\cot_{\theta min}}\right) \leq \nu f_{cd} \\ \quad \cot_{\theta min} \\ \text{else if } 2\tau_{Ed} \leq \nu f_{cd} \\ \quad \frac{\nu f_{cd}}{2\tau_{Ed}} + \sqrt{\left(\frac{\nu f_{cd}}{2\tau_{Ed}}\right)^2 - 1} \\ \text{else} \\ \quad 1 \end{cases} = 2.5$$

7.3 Resistance

Shear reinforcement ratio (8.43), set 2 when δ2 = 1

$$\rho_w := \frac{2}{b_w}\left(\frac{A_{w1}}{s_1} + \delta_2 \frac{A_{w2}}{s_2}\right) = 0.00722992$$

Spiral resistance (8.42)

$$\tau_{Rd.sy} := \rho_w f_{ywd} \cot_\theta = 7.86 \cdot \text{MPa}$$

Strut stress (8.44)

$$\sigma_{cd} := \tau_{Ed}\left(\cot_\theta + \frac{1}{\cot_\theta}\right) = 8.65 \cdot \text{MPa}$$

Shear utilisation ratio

$$\text{UR}_V := \max\left(\frac{\tau_{Ed}}{\tau_{Rd.sy}}, \frac{\sigma_{cd}}{\nu f_{cd}}\right) = 0.8718$$

Check $\mathrm{UR}_V = 0.8718 \leq 1$ PASS

8. SUMMARY

Flexure utilisation ratio $\mathrm{UR}_M = 0.7689$

Shear utilisation ratio $\mathrm{UR}_V = 0.8718$

ANNEX

A.1 Bar coordinates

Bar index, set 1 (edit seq to renumber) $i_1 := \mathrm{seq}\left(0, n_{l1} - 1\right)$

Bar index, set 2 $i_2 := \mathrm{seq}\left(0, n_{l2} - 1\right)$

Angles for drawing circles $\varphi_t := \mathrm{seq}\left(0, 2\pi, \frac{\pi}{36}\right)$

Set-1 bar coordinates, x $x_{l1} := r_1 \cos\left(\frac{2\pi i_1}{n_{l1}}\right)$

Set-1 bar coordinates, y $y_{l1} := r_1 \sin\left(\frac{2\pi i_1}{n_{l1}}\right)$

Set-2 bar coordinates, x $x_{l2} := r_2 \cos\left(\frac{2\pi i_2}{n_{l2}}\right)$

Set-2 bar coordinates, y $y_{l2} := r_2 \sin\left(\frac{2\pi i_2}{n_{l2}}\right)$

Set 1 (outer)

$$i_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \\ 15 \\ 16 \\ 17 \\ 18 \\ 19 \\ 20 \\ 21 \end{bmatrix} \quad x_{l1} = \begin{bmatrix} 489 \\ 469 \\ 411 \\ 320 \\ 203 \\ 70 \\ -70 \\ -203 \\ -320 \\ -411 \\ -469 \\ -489 \\ -469 \\ -411 \\ -320 \\ -203 \\ -70 \\ 70 \\ 203 \\ 320 \\ 411 \\ 469 \end{bmatrix} \text{ mm} \quad y_{l1} = \begin{bmatrix} 0 \\ 138 \\ 264 \\ 370 \\ 445 \\ 484 \\ 484 \\ 445 \\ 370 \\ 264 \\ 138 \\ 0 \\ -138 \\ -264 \\ -370 \\ -445 \\ -484 \\ -484 \\ -445 \\ -370 \\ -264 \\ -138 \end{bmatrix} \text{ mm}$$

Set 2 (inner, used when nset = 2)

$$i_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \\ 15 \end{bmatrix} \quad x_{l2} = \begin{bmatrix} 444 \\ 411 \\ 314 \\ 170 \\ 0 \\ -170 \\ -314 \\ -411 \\ -444 \\ -411 \\ -314 \\ -170 \\ 0 \\ 170 \\ 314 \\ 411 \end{bmatrix} \text{ mm} \quad y_{l2} = \begin{bmatrix} 0 \\ 170 \\ 314 \\ 411 \\ 444 \\ 411 \\ 314 \\ 170 \\ 0 \\ -170 \\ -314 \\ -411 \\ -444 \\ -411 \\ -314 \\ -170 \end{bmatrix} \text{ mm}$$

A.2 Concrete strips

The concrete is cut into nc thin strips parallel to the neutral axis. Each strip carries the stress at its centre strain; summing strip forces (× lever arm) gives the concrete N and M.

Strip index $j := \text{seq}(0, n_c - 1)$

Strip width $\Delta := \dfrac{D}{n_c} = 12 \cdot \text{mm}$

Strip centre coordinate, from the centroid $x_c := R - (j + 0.5)\,\Delta$

Strip area (chord length × width) $A_c := 2\sqrt{R^2 - x_c^2} \cdot \Delta$

Strip area / gross area (check of nc) $\dfrac{\text{sum}(A_c)}{A_g} = 1.0003$

A.3 Strain plane, stress laws and section resultants

Set-1 bar coordinate p along θM (from centroid)

$$p_{l1} := r_1 \cos\left(\frac{2\pi i_1}{n_{l1}}\,\text{rad} - \theta_M\right)$$

Set-2 bar coordinate p along θM

$$p_{l2} := r_2 \cos\left(\frac{2\pi i_2}{n_{l2}}\,\text{rad} - \theta_M\right)$$

Depth dθ: compression edge to extreme bar

$$d_\theta := \begin{cases} \text{if } n_{set} \geq 2 \\ \quad \| R - \min(p_{l1}, p_{l2}) \\ \text{else} \\ \quad \| R - \min(p_{l1}) \end{cases} = 1089 \cdot \text{mm}$$

Depth cA at pivot A/B boundary

$$c_A := \frac{\varepsilon_{cu2} d_\theta}{\varepsilon_{cu2} + \varepsilon_{ud}} = 79 \cdot \text{mm}$$

Depth yC of pivot C (εc2 in full compression)

$$y_C := \left(1 - \frac{\varepsilon_{c2}}{\varepsilon_{cu2}}\right) D = 514 \cdot \text{mm}$$

Strain plane (+ compression)

c = neutral-axis depth below the compression edge; p = fibre coordinate from the centroid along θM, so R − p = fibre depth. The plane pivots about A (extreme bar at −εud, c < cA), B (edge at εcu2, c ≤ D) or C (εc2 at depth yC, c > D), EC2 Fig. 6.1.

Strain at fibre p for depth c

$$\varepsilon(c, p) := \begin{cases} \text{if } c < c_A \\ \quad \left\| \varepsilon_{ud} \dfrac{c - R + p}{d_\theta - c} \right. \\ \text{else if } c \leq D \\ \quad \left\| \varepsilon_{cu2} \dfrac{c - R + p}{c} \right. \\ \text{else} \\ \quad \left\| \varepsilon_{c2} \dfrac{c - R + p}{c - y_C} \right. \end{cases}$$

Stress laws

Concrete stress, parabola-rectangle (EC2 3.17), no tension

$$\sigma_c(\varepsilon) := \begin{cases} \text{if } \varepsilon \le 0 \\ \quad 0 \cdot \text{MPa} \\ \text{else if } \varepsilon < \varepsilon_{c2} \\ \quad f_{cd}\left(1 - \left(1 - \frac{\varepsilon}{\varepsilon_{c2}}\right)^2\right) \\ \text{else} \\ \quad f_{cd} \end{cases}$$

Steel stress, elastic-plastic within ±fyd

$$\sigma_s(\varepsilon) := \begin{cases} \text{if } E_s\varepsilon < -f_{yd} \\ \quad -f_{yd} \\ \text{else if } E_s\varepsilon > f_{yd} \\ \quad f_{yd} \\ \text{else} \\ \quad E_s\varepsilon \end{cases}$$

Net bar stress (concrete it displaces subtracted)

$$\sigma_b(\varepsilon) := \sigma_s(\varepsilon) - \sigma_c(\varepsilon)$$

Section resultants

Concrete strips (xc, Ac) plus bars of set 1 and, when δ2 = 1, set 2:

Axial resistance for depth c

$$N_R(c) := \text{sum}\left(\sigma_c\left(\varepsilon\left(c, x_c\right)\right) A_c\right) + A_{b1}\text{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l1}\right)\right)\right) + \delta_2 A_{b2}\text{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l2}\right)\right)\right)$$

Moment resistance for depth c (about the centroid)

$$M_R(c) := \text{sum}\left(\sigma_c\left(\varepsilon\left(c, x_c\right)\right) A_c x_c\right) + A_{b1}\text{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l1}\right)\right) p_{l1}\right) + \delta_2 A_{b2}\text{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l2}\right)\right) p_{l2}\right)$$

A.4 P–M diagram data (control points)

Neutral-axis depths: −0.1D (all bars yielded in tension) … 100D (uniform εc2)

$$c_v := \text{stack}\left(\text{seq}\left(-0.1, 1, 0.025\right) D, \frac{D}{\text{seq}\left(0.9, 0.1, -0.1\right)}, 100D\right)$$

Number of control points

$$\text{len}\left(c_v\right) = 55$$

Strain at the compression edge for each point $\varepsilon_{top} := \mathrm{map}\left(\varepsilon, c_v, R\right)$

Strain at the extreme tension bar for each point $\varepsilon_{bar} := \mathrm{map}\left(\varepsilon, c_v, R - d_\theta\right)$

Axial resistance at each point $N_v := \mathrm{map}\left(N_R, c_v\right)$

Moment resistance at each point $M_v := \mathrm{map}\left(M_R, c_v\right)$

Control points at θM (N compression +)

$$c_v = \begin{matrix}\text{rows 1:28}\\ \begin{bmatrix}-120\\-90\\-60\\-30\\0\\30\\60\\90\\120\\150\\180\\210\\240\\270\\300\\330\\360\\390\\420\\450\\480\\510\\540\\570\\600\\630\\660\\690\end{bmatrix}\end{matrix} \quad \begin{matrix}\text{rows 29:55}\\ \begin{bmatrix}720\\750\\780\\810\\840\\870\\900\\930\\960\\990\\1020\\1050\\1080\\1110\\1140\\1170\\1200\\1333\\1500\\1714\\2000\\2400\\3000\\4000\\6000\\12000\\120000\end{bmatrix}\end{matrix} \ \mathrm{mm}$$

$$\varepsilon_{top} = \begin{matrix} \text{rows } 1{:}28 \\ \begin{bmatrix} -0.0044668 \\ -0.0034353 \\ -0.00235 \\ -0.0012065 \\ 0 \\ 0.0012749 \\ 0.0026241 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \end{bmatrix} \end{matrix} \quad \begin{matrix} \text{rows } 29{:}55 \\ \begin{bmatrix} 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0035 \\ 0.0032558 \\ 0.0030435 \\ 0.0028571 \\ 0.0026923 \\ 0.0025455 \\ 0.0024138 \\ 0.0022951 \\ 0.0021875 \\ 0.0020896 \\ 0.0020086 \end{bmatrix} \end{matrix}$$

$$\varepsilon_{bar} = \begin{matrix} \text{rows } 1{:}28 \\ \begin{bmatrix} -0.045 \\ -0.045 \\ -0.045 \\ -0.045 \\ -0.045 \\ -0.045 \\ -0.045 \\ -0.038847 \\ -0.02826 \\ -0.021908 \\ -0.017674 \\ -0.014649 \\ -0.01238 \\ -0.010616 \\ -0.0092042 \\ -0.0080493 \\ -0.0070868 \\ -0.0062725 \\ -0.0055744 \\ -0.0049695 \\ -0.0044401 \\ -0.0039731 \\ -0.0035579 \\ -0.0031864 \\ -0.0028521 \\ -0.0025496 \\ -0.0022746 \\ -0.0020236 \end{bmatrix} \end{matrix} \quad \begin{matrix} \text{rows } 29{:}55 \\ \begin{bmatrix} -0.0017934 \\ -0.0015817 \\ -0.0013862 \\ -0.0012053 \\ -0.0010372 \\ -0.00088076 \\ -0.00073473 \\ -0.00059813 \\ -0.00047006 \\ -0.00034976 \\ -0.00023653 \\ -0.00012977 \\ -2.8943 \times 10^{-5} \\ 6.6434 \times 10^{-5} \\ 0.00015679 \\ 0.00024251 \\ 0.00032395 \\ 0.0005968 \\ 0.00083405 \\ 0.0010423 \\ 0.0012264 \\ 0.0013905 \\ 0.0015376 \\ 0.0016703 \\ 0.0017905 \\ 0.0018999 \\ 0.0019904 \end{bmatrix} \end{matrix}$$

$N_v =$ rows 1:28 $\begin{bmatrix} -7693 \\ -7693 \\ -7693 \\ -7693 \\ -7693 \\ -7631 \\ -7410 \\ -6725 \\ -5783 \\ -4771 \\ -3868 \\ -2911 \\ -2061 \\ -1122 \\ -236 \\ 587 \\ 1486 \\ 2317 \\ 3147 \\ 4032 \\ 4889 \\ 5754 \\ 6632 \\ 7483 \\ 8396 \\ 9294 \\ 10239 \\ 11245 \end{bmatrix}$ rows 29:55 $\begin{bmatrix} 12235 \\ 13180 \\ 14091 \\ 14969 \\ 15821 \\ 16647 \\ 17451 \\ 18229 \\ 18984 \\ 19714 \\ 20423 \\ 21099 \\ 21747 \\ 22363 \\ 22952 \\ 23511 \\ 24037 \\ 25364 \\ 26375 \\ 27153 \\ 27751 \\ 28210 \\ 28568 \\ 28834 \\ 29002 \\ 29100 \\ 29159 \end{bmatrix}$ kN

$M_v =$ rows 1:28 $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 36 \\ 162 \\ 507 \\ 962 \\ 1424 \\ 1821 \\ 2212 \\ 2552 \\ 2885 \\ 3186 \\ 3448 \\ 3689 \\ 3899 \\ 4089 \\ 4253 \\ 4402 \\ 4520 \\ 4608 \\ 4679 \\ 4712 \\ 4727 \\ 4702 \\ 4622 \end{bmatrix}$ rows 29:55 $\begin{bmatrix} 4528 \\ 4426 \\ 4315 \\ 4197 \\ 4072 \\ 3940 \\ 3799 \\ 3652 \\ 3497 \\ 3337 \\ 3169 \\ 2997 \\ 2820 \\ 2641 \\ 2461 \\ 2282 \\ 2108 \\ 1608 \\ 1228 \\ 935 \\ 708 \\ 530 \\ 391 \\ 278 \\ 174 \\ 78 \\ 7 \end{bmatrix}$ kN · m

Max. axial resistance (uniform εc2) $N_{Rd.max} := \max(N_v) = 29159.2 \cdot \text{kN}$

Maximum moment resistance on the curve $M_{Rd.max} := \max(M_v) = 4726.7 \cdot \text{kN} \cdot \text{m}$

Normalised N of the control points

$$n_v := \frac{N_v}{N_{Rd.max}}$$

Normalised M of the control points

$$m_v := \frac{M_v}{M_{Rd.max}}$$

Distance of each control point from the origin

$$\rho_v := \sqrt{n_v^2 + m_v^2}$$

Direction of each point (sine of angle)

$$s_v := \frac{n_v}{\rho_v}$$